ASUx48 - Scientific Research
Assignment 2 - Literature Scan

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Research Question
What is the role of IoT in enhancing automation and efficiency in smart building systems?

List of Articles

#	Article Name	Year	Article Type	Authors	Importance	Link		Keywords	Relevance
						Full Text	EKB		
1	A Review of Using IoT for Energy Efficient Buildings and Cities: A Built Environment Perspective	2022	Review	Karam M. Al-Obaidi Mohataz Hossain Nayef A. M. Alduais Husam S. Al-Duais Hossein Omrany Amirhosein Ghaffarianhoseini	1	https://www.mdpi.com/1996-1073/15/16/5991	https://08105kd5y-1104-y-https-www-scopus-com.mplbci.ekb.eg/pages/publications/85137725538	IoT, Energy Efficiency, Smart Buildings, Urban Systems	From the Paper's abstract it is cleat that this paper gives a comprehensive overview of how IoT improves energy efficiency in smart buildings and urban settings. This will introduces the big picture and gives me the background knowledge I need before diving into more specific research.
2	IoT—A Promising Solution to Energy Management in Smart Buildings: A Systematic Review, Applications, Barriers, and Future Scope	2024	Review	Mukilan Poyyamozhi Balasubramanian Murugesan Narayanamoorthi Rajamanickam Mohammad Shorfuzzaman Yasser Aboelmagd	1	https://www.mdpi.com/2075-5309/14/11/3446	https://08105kd5y-1104-y-https-www-scopus-com.mplbci.ekb.eg/pages/publications/85210240978	IoT, Smart Buildings, Energy Management, Barriers, Applications	This review focuses on practical applications of IoT for energy management in smart buildings and offers a deep dive into key challenges such as high initial costs, data security, and integration complexity. Unlike general overviews, it emphasizes real-world adoption barriers and solutions.

3	Smart Buildings: A Comprehensive Systematic Literature Review on Data-Driven Building Management Systems	2024	Review	Adrian Taboada-Orozco Kokou Yetongnon Christophe Nicolle	2	https://www.mdpi.com/1424-8220/24/13/4405	https://08105kd5y-1104-y-https-www-scopus-com.mplbci.ekb.eg/pages/publications/85198392602	Smart Buildings, IoT Architecture, Building Management Systems (BMS), Literature Mapping	This article offers a structured mapping of smart building research, categorizing systems, technologies, and research trends. It emphasizes the role of IoT architectures and BMS platforms, making it useful for understanding how different smart systems integrate within building environments.
4	IoT-Based Real-Time Monitoring System for a Smart Energy House	2021	Research	Lukasz Rojek Saiful Islam Michael Hartmann Reiner Creutzburg	1	https://www.researchgate.net/publication/354766082_IoT-Based_Real-Time_Monitoring_System_for_a_Smart_Energy_House	https://08105kd9g-1104-y-https-www-scopus-com.mplbci.ekb.eg/pages/publications/105008959478	IoT, Smart Energy House, Monitoring System, MQTT	This study presents a smart monitoring system for a building that tracks conditions like temperature, water levels, and battery usage in real time. It shows how Internet of Things (IoT) devices can work together to improve comfort, energy use, and safety in homes. The system allows both local and remote access, making it practical and adaptable for real-world use.
5	Smart Buildings and Digital Twin to Monitoring the Efficiency and Wellness of Working Environments: A Case Study on IoT Integration and Data-Driven Managemen	2025	Research	Giuseppe Piras Sofia Agostinelli Francesco Muzi	2	https://www.mdpi.com/2076-3417/15/9/4939	https://08105kd9g-1104-y-https-www-scopus-com.mplbci.ekb.eg/pages/publications/105004880414	Smart Buildings, Digital Twin, IoT, Energy Efficiency, Indoor Comfort	This article presents a working system that uses IoT and Digital Twin technology to monitor and improve energy use, air quality, lighting, and comfort in a smart workplace. It includes a case study with real data collection and shows how dashboards and ML algorithms support better decision-making for both energy efficiency and employee well-being.
6	Technological fascination and reluctance: gendered practices in the smart home	2022	Research	Aagaard Line Kryger Madsen Line Valdorff	3	https://journal-buildingscities.org/articles/10.5334/bc.205	https://08105kd9g-1104-y-https-www-scopus-com.mplbci.ekb.eg/pages/publications/85139181310	Gender, Smart Home, User Experience, Technology Adoption, Behavior	This study explores how men and women interact differently with smart home technologies in everyday life. It shows that digital housekeeping tasks (like using apps or controlling devices) are often handled by men, while traditional household tasks are still done by women. These differences create digital inequality in smart homes. The paper highlights the importance of designing smart technologies that consider the skills, roles, and needs of all users — not just the technically confident ones.
7	The Main Barriers Limiting the Development of Smart Buildings	2024	Research	Affonso, Estefany O. T. Branco, Robson R. Menezes, Osvaldo V. C. Guedes, André L. A. Chinelli, Christine K.	1	https://www.mdpi.com/2075-5309/14/6/1726	https://08105kddd-1104-y-https-www-scopus-com.mplbci.ekb.eg/pages/publications/85197251800	Smart Buildings, Barriers, Construction, Policy, Cost, Professionals	This article identifies and ranks 23 barriers that slow down the growth of smart buildings, such as lack of skilled professionals, high technology costs, weak government support, and limited awareness of benefits. It is based on literature review and expert opinions, making it a strong reference for understanding practical challenges in real-world adoption.

8	Privacy for IoT: Informed consent management in Smart Buildings	2023	Research	Chehara Pathmabandu John Grundy Mohan Baruwal Chhetri Zubair Baig	2	https://www.sciencedirect.com/science/article/pii/S0167739X23001322	Privacy for IoT: Informed consent management in Smart Buildings	Privacy, IoT, Smart Buildings, Consent Management, User Awareness	This study addresses the tension between convenience and privacy in smart buildings by proposing an Informed Consent Management Engine (ICME). It raises awareness about data collection in IoT-enabled buildings and allows users to make more informed privacy decisions. The paper contributes a new dimension to smart building systems by emphasizing ethical design, transparency, and user empowerment.
9	IoT-based Occupancy Monitoring Techniques for Energy-Efficient Smart Buildings	2015	Research	Kemal Akkaya Ismail Guvenc Ramazan Aygun Nezih Pala Abdullah Kadri	3	https://www.researchgate.net/publication/280157569_IoT-based_Occupancy_Monitoring_Techniques_for_Energy-Efficient_Smart_Buildings	https://08105kd9g-1104-y-https-www-scopus-com.mplbci.ekb.eg/pages/publications/84938811184	IoT, Occupancy Monitoring, Energy Efficiency, Smart HVAC, Data Fusion	This article reviews and compares different IoT-based occupancy tracking methods used in commercial buildings. It highlights how combining data from sensors, WiFi, and RFIDs helps automate HVAC systems in real time, reducing energy waste. It also discusses challenges in privacy, accuracy, and infrastructure use, making it a useful reference for automation strategies in smart buildings.
10	Internet of Things (IoT) as a Tool in Architecture for The Concept of Sustainability in Egypt	2022	Research	Vitta Abdel Rehim Ibrahim		https://sej.journals.ekb.eg/article_261147.html	-	IoT, Smart Buildings, Architecture, Egypt, Energy Efficiency, Sustainability	This paper presents a real-world case study of IoT adoption in administrative buildings within the Beverly Hills Compound, Sheikh Zayed, Egypt. It uses a survey-based approach to show how IoT enhances energy efficiency and supports sustainable design. Its accessible, non-technical presentation makes it valuable for understanding local applications of smart building technologies.
11	Fast Forwards: 10 Years of Sustainable Initiatives in the Gulf Region	2015	Review	Willis Jeffrey	1	https://sci-hub.in/10.1002/ad.1861	https://08105kd9g-1104-y-https-www-scopus-com.mplbci.ekb.eg/pages/publications/85208201122	Applications - Al Bahr Towers - Gulf Region	This article reviews sustainable architecture efforts in the Gulf region, highlighting major projects and green building policies such as Estidama, GSAS, and LEED. It features **Al Bahr Towers** in Abu Dhabi as a key example, showcasing the use of automated **mashrabiya** façades to reduce solar gain, lower energy consumption, and promote environmentally friendly design. The article provides a valuable real-world case of smart and green building implementation in the Middle East.
12	Occupancy Prediction in IoT-Enabled Smart Buildings: Technologies, Methods, and Future Directions	2024	Review	Irfanullah Khan Ouarda Zedadra Antonio Guerrieri Giandomenico Spezzano	3	https://www.mdpi.com/1424-8220/24/11/3276	https://08105kd9g-1104-y-https-www-scopus-com.mplbci.ekb.eg/pages/publications/85195831944	Occupancy Prediction, IoT, Smart Buildings, Energy Efficiency, Sensor Placement	This article reviews modern IoT-based techniques used to monitor and predict occupant presence in buildings. It explains how occupancy prediction helps buildings automate HVAC and lighting systems, reducing energy waste. The review also discusses challenges in deploying these systems and offers future directions for smarter, more efficient building environments.